Digital Thorium

A Market Architecture for India's Thorium Century

The Convergence of Reserves, Compute, and Ownership

This report proposes that India create a market for tradable, fractional claims on thorium-derived electricity. It argues the case sits at the intersection of three facts: India holds roughly a quarter of the world's thorium; artificial intelligence is about to become the largest new consumer of electrons on the planet; and in the coming decade more people will want a stake in energy than in land, because owning energy increasingly means owning intelligence. The document lays out the protocol stack, runs the economic simulation, and shows why a market based approach serves the national interest, the security interest, and every identifiable stakeholder better than state monopoly, pure private issuance, or pure renewable build-out.

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A zero to frontier policy report

Abhilash Inumella · *Anubhav Tiwari*

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Executive Summary

India is sitting on the largest thorium deposit in the world. It is not using it. Meanwhile the global price of one unit of reliable, clean electricity is about to become the price of one unit of intelligence. This report asks a simple question: who should own the upside when India finally turns rock into power into cognition?

The answer proposed here is: ordinary Indians, priced by a market, secured by the state. The instrument is a tradable, fractional, on-chain claim on a unit of thorium-derived electricity produced by a federally supervised reactor fleet. Call it **Digital Thorium**. One token, one kilowatt-hour-year of claim, with redemption, secondary trading, and derivative markets governed by a new authority sitting between the Department of Atomic Energy and the Securities and Exchange Board of India.

The thesis rests on three convergences:

1. **Reserves.** India holds an estimated 846,477 tonnes of thorium metal, roughly 25 percent of the known global inventory. The Prototype Fast Breeder Reactor at Kalpakkam achieved first criticality on 6 April 2026, closing the physics gap on the three-stage programme conceived by Homi Bhabha in 1954.
2. **Demand.** The International Energy Agency projects data centre electricity demand to more than double from 415 TWh in 2024 to 945 TWh by 2030 in the base case, and to nearly 2,000 TWh by 2035 in the lift-off case. AI-optimised load is projected to quadruple in the same window. Electrons are the new oil and compute is the new refinery.
3. **Ownership.** Land is a nineteenth century asset. In the coming decade more people will want a stake in energy than in land, because the marginal value of an additional kilowatt now accrues to whoever controls the training run, the inference workload, and the agent. Owning energy is owning intelligence.

The market-based design delivers five outcomes a state programme cannot: (i) capital formation at civilisational scale through retail participation; (ii) continuous price discovery that makes the true cost of clean baseload visible; (iii) a household balance sheet hedge against the inflationary pressure of an AI-driven electricity spike; (iv) a non-military instrument of strategic autonomy; and (v) a credible commitment device that survives political cycles.

The national interest is served because Digital Thorium turns the most abundant domestic resource into a financeable asset class. The security interest is served because thorium is proliferation resistant, the build-out is sovereign, and compute sovereignty requires electron sovereignty. Every major stakeholder, from the retail saver to the global hyperscaler, gets a better deal than under the current architecture.

CHAPTER 1

The Three Convergences

1.1 Resource Asymmetry: The Thorium Base

The Atomic Minerals Directorate for Exploration and Research places India's in-situ monazite reserve at 11.93 million tonnes, containing approximately 963,000 tonnes of thorium oxide, equivalent to 846,477 tonnes of thorium metal. This puts India at roughly 25 percent of the global inventory, tied with Brazil and ahead of the United States.

The distribution is coastal. The three eastern states of Andhra Pradesh, Tamil Nadu, and Odisha hold 72 percent of the reserve. Kerala holds a further slice. The resource is a placer deposit, meaning open-pit friendly and geopolitically concentrated in territory India already administers. There are no external choke points, no offshore disputes, no foreign equity.

Thorium cannot be split directly. It is a fertile isotope. When a thorium-232 nucleus absorbs a neutron it becomes protactinium-233, which beta decays to uranium-233, which is fissile. The three-stage nuclear programme designed by Homi Bhabha in the mid-1950s is the physical pipeline for this transition:

1. **Stage 1:** Pressurised heavy water reactors burn natural uranium and breed plutonium-239. India operates 20 such units with average capacity factors around 80 percent.
2. **Stage 2:** Fast breeder reactors use the accumulated plutonium to breed more fissile material while wrapping the core in a thorium blanket that generates U-233. The 500 MWe Prototype Fast Breeder Reactor at Kalpakkam attained first criticality on 6 April 2026.
3. **Stage 3:** Thorium-U233 reactors come online, converting the U-233 inventory into civilisational-scale electricity.

On the Manmohan Singh 2009 estimate, the full programme could deliver 470 GW by 2050. Department of Atomic Energy scientists put the economically extractable resource at 500 GWe for four centuries. For comparison, India's total installed capacity across all sources stood at 427 GW in 2025.

The physics is no longer the bottleneck. The bottleneck is capital, regulation, and public legitimacy. That is a market design problem, not an engineering problem.

1.2 Demand Shock: The Compute Hunger

The second convergence is that the electron is becoming the most strategically scarce commodity on earth. The IEA's April 2025 *Energy and AI* report projects:

Metric	2024	2030 Base	2035 Lift-Off
Global data centre demand (TWh)	415	945	∼2,000
Share of global electricity (%)	1.5	3.0	4.4
AI-accelerated server CAGR (%)	–	30	30+
Projected US DC demand growth (TWh)	–	+240	+400
Projected China DC demand growth (TWh)	–	+175	+350

Three features of this demand profile matter for India:

First, it is **baseload**. Training runs and inference workloads prefer round-the-clock electrons. Intermittent renewables alone cannot serve it without storage, and storage at this scale is neither cheap nor sovereign.

Second, it is **concentrated**. Data centres cluster in places with cheap, firm, long-duration power. The country that offers that becomes the compute residence of the world. The country that does not becomes the customer.

Third, it is **geopolitical**. The United States and China between them account for almost 80 percent of projected growth. India is absent from the top tier. Without a decisive supply-side move, Indian AI workloads will be trained in Virginia and Guizhou, billed in dollars and renminbi, and shaped by foreign jurisdictions.

India's current electricity mix is 79 percent coal-dominated. That is not a base you can credibly sell to a hyperscaler signing a twenty-year power purchase agreement. Thorium, by contrast, is clean, domestic, and effectively inexhaustible on the relevant time horizon.

1.3 Ownership Shift: Energy as the New Land

The third convergence is social. It concerns what a middle class saver in Patna or Pune wants to own in 2030.

For two centuries, land was the default store of value in India. It was legible, local, inheritable, and appreciated. But land is geographically fixed, low yielding in cash terms, vulnerable to political discretion, and impossible to fractionalise below a plot. Its returns are also largely uncorrelated with the coming productivity wave.

Energy, specifically clean baseload energy, is the opposite. It is fungible, priced in real time, yield-bearing through power purchase agreements, and tightly correlated with the productivity of the AI economy. If a household owns a claim on 10,000 kWh-years of thorium output, that claim benefits directly as the marginal hour of GPU time rises in price.

There is a philosophical framing worth naming. Georg Simmel argued in *Philosophy of Money* that money is crystallised social relation, and that the instruments a society builds to represent value shape its social form. What a culture finances, it eventually becomes. If India finances more land speculation, it gets a rentier economy. If it finances thorium, it gets a compute economy. The asset chosen is the future chosen.

The prediction: by 2035, more Indians will hold a claim on a kilowatt than on a katha. The reason is simple. A kilowatt pays you. A katha does not.

CHAPTER 2

What Digital Thorium Is

2.1 The Instrument

Digital Thorium is a tradable, fractional, on-chain claim on a defined quantity of electricity produced by a licensed thorium reactor operating on Indian territory under Department of Atomic Energy supervision.

The unit of account is one **kilowatt-hour-year** (kWhy): a claim on the delivery, or the monetary equivalent at settlement price, of one kWh per year, for the contract duration, from a specified reactor cohort. Think of it as a zero-coupon delivery contract with a perpetual or multi-decade tenor.

The issuance is federal. The issuer is a new vehicle, call it the **National Thorium Finance Corporation** (NTFC), sitting inside the Department of Atomic Energy ecosystem. Reactors are built and operated by NPCIL, BHAVINI, and licensed private consortia. Each reactor cohort, once certified, is tokenised against a verified nameplate capacity and an expected capacity factor.

Holders have three rights:

1. **Cash settlement** at the prevailing wholesale power price, paid quarterly in rupees.
2. **Physical take-up** above a threshold size, for industrial buyers who want behind-the-meter or direct PPA service.
3. **Secondary trading** on a regulated exchange with continuous price discovery.

The architecture is intentionally post-crypto. The token is the database entry. The settlement is rupee. The enforcement is Indian law. The promise is electricity. The speculative complexity belongs in the derivatives layer, not the base layer.

2.2 The Protocol Stack

A clean mental model separates four layers, each with its own regulator, its own economics, and its own failure mode.

Layer	**Function**	**Primary Regulator**	**Failure Mode**
Physical	Thorium extraction, fuel fabrication, reactor ops	AERB, DAE	Capacity shortfall
Settlement	PPA enforcement, rupee payment, grid metering	CERC, POSOCO	Price suppression
Token	Issuance, custody, transfer, redemption	SEBI, NTFC	Counterparty fraud
Derivative	Futures, options, structured products	SEBI, RBI	Speculative blowout

The physical layer is the bedrock. No token is issued without commissioned nameplate capacity. This is the opposite of crypto-economic bootstrapping: the real asset must exist before the financial claim.

The settlement layer ensures that the promise of a kilowatt-hour is real. It uses existing electricity market infrastructure: the Central Electricity Regulatory Commission, the Power System Operation Corporation, and the two operating exchanges (IEX and PXIL). The NTFC sits in the middle as the aggregator and defaulter-of-last-resort.

The token layer is a permissioned ledger, not a public chain. The design goal is legal certainty, not censorship resistance. Transfers require KYC. Foreign ownership is capped and tagged. The ledger is auditable by Parliament.

The derivative layer is where speculation lives. Futures let industrial buyers hedge power costs. Options let retail savers take or sell upside. Structured products let pension funds match long-dated liabilities. This is where most of the liquidity accumulates, and where most of the risk must be contained.

CHAPTER 3

Economic Analysis: Why a Market Beats a Monopoly

3.1 The Core Argument

The orthodox case for state monopoly in nuclear power rests on three claims: safety, non-proliferation, and capital scale. All three are real. None of them imply financial monopoly.

A safe reactor does not care who owns the kilowatt-hour it produces. A proliferation-resistant fuel cycle does not become less secure because a retail saver in Guntur holds a claim on its output. A capital-intensive project does not become cheaper to finance by narrowing the investor base.

Indeed the opposite. The fundamental theorem of finance is that the cost of capital falls as the risk pool widens. A state monopoly forces the Union Budget to absorb the full variance of a 500 GW build-out. A market distributes that variance across the savings of 400 million households, the balance sheets of pension funds, and the strategic allocations of sovereign partners. Everyone pays less for the same reactor.

The market mechanism also produces a price. The budget does not. Under the current regime nobody knows what a clean Indian kilowatt-hour is really worth in 2045, because no instrument quotes it. Under Digital Thorium, the market quotes it every day, and the quote feeds every downstream decision: which data centre to build, which PPA to sign, which reactor cohort to green-light next.

3.2 Capital Formation Simulation

To make the economics concrete, consider a 50 GW thorium-cycle build-out over twenty years. This is roughly one tenth of the long-run potential estimated by DAE, and broadly consistent with the 100 GW target touted in current policy conversation, but built through a market channel.

Assumptions.

Parameter	**Value**
Target nameplate capacity by 2046	50,000 MW
Overnight capital cost per MW (FOAK $\rightarrow$ NOAK blended)	\$5.0M
Total construction capex (2026 \$)	\$250B
Build period	20 years
Capacity factor (steady state)	82%
Expected plant life	60 years
Target levelised cost of electricity (LCOE)	INR 4.5–6.0/kWh
Equity/debt split at project level	30/70
Digital Thorium token share of project equity	40%
Retail share of token float (target)	50%

At an 82 percent capacity factor a 50 GW fleet produces approximately $50{,}000 \times 8760 \times 0.82 = 359$ TWh per year. That is about 21 percent of India's 2025 generation and enough to power a data centre footprint equivalent to the entire current American fleet at 2024 utilisation.

Issuance sizing. Total tokenisable equity is $0.40 \times 0.30 \times \$250\text{B} = \$30$ billion, or roughly INR 2.5 lakh crore over 20 years. At a $1kWhyunit, roughly360billionkWhyclaimscanbecarvedfromtheannualoutput.At$

Household reach. If 50 percent of the token float is reserved for retail at a minimum ticket of INR 500 (the same as a mutual fund SIP), the structure can onboard of the order of 200 million households without any single ticket exceeding the savings rate of a median family. This is comparable in scale to the Jan Dhan account rollout, and uses the same India Stack rails.

Capital formation velocity. Under a state monopoly, India's nuclear build-out has averaged roughly 0.3 GW per year over the last two decades. Under a market channel with 200 million retail participants and a cost of equity reduced by 300 basis points, the annual build rate can credibly target 2.5 GW, an eight-fold acceleration. The arithmetic: lowering the cost of equity from 14 percent to 11 percent on a 60-year asset compresses the LCOE by roughly 25 percent, which widens the band of commercially viable projects by more than a factor of two.

3.3 Price Discovery as National Infrastructure

A market does something a monopoly cannot: it produces a continuous, public, litigable quote.

Today nobody in India can tell you with confidence what a kilowatt-hour of clean baseload will cost in 2045. The Central Electricity Authority publishes projections. The Power Ministry publishes tariff orders. NITI Aayog publishes scenarios. All of them disagree. None of them is traded.

Digital Thorium replaces this opacity with a price. The price aggregates the beliefs of every participant: the hyperscaler hedging GPU power costs, the pension fund matching a liability in 2055, the retail saver making a ten-year bet on India's AI trajectory. The quote updates every tick. That quote then feeds every downstream allocation decision in the economy.

The informational externality is large. A credible forward curve for Indian clean baseload:

- Tells hyperscalers where to build data centres
- Tells startups whether compute-heavy business models are viable
- Tells the Finance Ministry how much dollar inflow to expect
- Tells state governments which land parcels to industrialise
- Tells the RBI how to model the energy component of future inflation

None of this information exists today at actionable granularity. Digital Thorium manufactures it.

3.4 Comparative Vehicles

It is worth being explicit about how Digital Thorium differs from existing ways of packaging energy cash flows.

Vehicle	What You Own	What You Miss	Why Different
NTPC equity	Claim on a coal-heavy mixed fleet	No thorium exposure, no commodity tracking	Corporate, not asset-linked
InvIT	Claim on specific transmission assets	No generation economics, regulated returns	No upside to AI demand
Green Bond	Fixed coupon on clean project	Paid in rupees, no price exposure	Debt, not equity in energy
REIT	Claim on physical real estate	Tied to land not electrons	Wrong convergence
Carbon credit	Claim on avoided emissions	Political durability uncertain	Not an energy claim
Digital Thorium	Claim on kWh-years of thorium output	Nothing structural	The missing instrument

The gap is real. Every existing vehicle is either a claim on a company, a claim on land, or a claim on a fixed cash flow. None of them give a retail saver direct exposure to the price of clean baseload electricity in the country where thorium is most abundant. Digital Thorium is that instrument.

3.5 What a Market Does That a Budget Cannot

Five things:

1. **Capital at scale without sovereign ceiling.** The Union Budget is a rival-good. Every rupee spent on thorium is a rupee not spent on roads, health, or defence. A market draws on savings that are already in the system and currently sitting in gold and real estate.

2. **Survives the electoral cycle.** A budget allocation can be reversed in the next Finance Act. Securitised claims held by 200 million voters cannot.

3. **Prices externalities.** Coal is subsidised by the atmosphere. Thorium is subsidised by no one. A market forces the externality into the quote.

4. **Disciplines the operator.** If NPCIL misses a capacity factor target, the token price drops and the NTFC is accountable. Under pure state ownership, there is nobody to punish.

5. **Exports itself.** Once the rails exist, they can list foreign thorium assets too. India becomes the price-setter, not the price-taker, in the global thorium market.

CHAPTER 4

The National Interest

4.1 From Resource Sovereignty to Intelligence Sovereignty

The National Interest test for any major policy is crude but useful: does this measurably increase India's room to manoeuvre in the next forty years? Digital Thorium scores on four axes.

Energy independence. India imported $158 billion of crude in FY24. Every kilowatt-hour delivered by a domestic thorium reactor substitutes for a marginal barrel. A 50 GW thorium fleet displaces oil imports of roughly $20–25 billion per year at plausible elasticities, before counting the secondary benefit of coal retirement.

Compute independence. The training of frontier AI models is already a sovereign act in the United States, China, and France. India cannot train frontier models on foreign soil and expect to own the resulting intelligence. Sovereign compute requires sovereign electrons. Digital Thorium is the financing channel.

Fiscal independence. A mature market for Digital Thorium supplies a long-duration, non-inflationary source of domestic savings for infrastructure. It reduces the need for foreign borrowing and lowers the sovereign spread.

Technological independence. The domestic supply chain for thorium is almost entirely onshore: monazite extraction in Kerala and Odisha, fuel fabrication at NFC Hyderabad, reactor design by BARC and IGCAR. There is no equivalent domestic chain for solar PV or lithium-ion batteries, both of which are Chinese-dominated. Digital Thorium tilts the energy transition towards technologies India actually owns.

4.2 The Demographic Dividend Trap and the Escape

India's demographic dividend closes around 2045. The conventional narrative says the country must industrialise before then. The inconvenient truth is that the world is deindustrialising labour. What matters is not manufacturing jobs but ownership of productive capital during the automation transition.

Energy ownership is the highest-leverage form of such capital. A household that owns 10,000 kWhy of thorium claims in 2030 owns a share of the compute output of the 2045 economy. This converts the demographic dividend from a labour dividend, which is depreciating, into a capital dividend, which is not.

This is a deliberately populist framing. It is also correct. The social return on a 500-rupee Digital Thorium SIP dwarfs the social return on the equivalent allocation to land or gold, and it aligns household prosperity with national technological ambition rather than setting them in tension.

4.3 The Fiscal Arithmetic

Fiscal effect (annualised, steady state)	**INR crore**
Oil import substitution (50 GW displacing refined liquid and coal)	1.6–2.0 lakh
Additional corporate tax from domestic compute sector	40–60 thousand
GST on hyperscaler energy contracts on Indian soil	25–40 thousand
Sovereign spread compression (50 bps on INR 70 lakh crore debt)	35 thousand
Gross fiscal benefit (rough, indicative)	$\sim$3.0–3.5 lakh
Government equity contribution required (over 20 years)	2.5 lakh

The policy pays for itself within roughly a decade of full ramp, before counting any of the strategic benefits. The residual is a permanent uplift to national income.

CHAPTER 5

The Security Interest

5.1 Non-Proliferation by Chemistry

Thorium has an unusual property from a security standpoint. Its use generates uranium-232 as a contaminant, which decays with an intense gamma signature. This makes the U-233 fuel cycle extremely difficult to weaponise covertly. Centrifuge enrichment, the classic proliferation pathway for U-235, is irrelevant for a thorium cycle. The International Panel on Fissile Materials has long noted that a thorium-U233 fuel cycle is materially harder to divert than an enriched uranium cycle.

This matters for the international optics of Digital Thorium. A state offering a large-scale civilian nuclear asset to retail investors would trigger, under a uranium cycle, immediate IAEA and NSG concerns. Under a thorium cycle, those concerns are substantially attenuated. India can scale the programme without re-litigating its 2008 waiver.

5.2 Compute Sovereignty

Modern national security is measured in tokens per second on domestic silicon drawing domestic electrons. Without sovereign compute, a country cannot:

- Train defence-relevant AI models on classified data
- Guarantee the uptime of mission critical inference in a crisis
- Set the terms on which its citizens' data is processed
- Prevent a foreign hyperscaler from becoming a de facto regulator of domestic speech and commerce

Sovereign compute has two ingredients: silicon and electrons. India is far behind on silicon. It does not need to be behind on electrons. Thorium at scale closes the electron gap decisively. Digital Thorium finances the closing.

5.3 Grid Hardening and Baseload Resilience

Renewables are essential. They are also attackable. A solar farm is easier to damage than a reactor. A grid whose firming depends on imported lithium is more fragile than one firmed by domestic thorium. Baseload nuclear, distributed across coastal clusters, with sovereign fuel supply, is the most resilient configuration available under any realistic threat model.

In the specific scenario of extended adversarial action against sea lanes, India's current energy mix is catastrophically exposed. LNG import disruption, crude import disruption, or coking coal import disruption would each, individually, force rationing within weeks. Thorium has no such exposure. The fuel is Indian. The enrichment is unnecessary. The cycle is closed.

5.4 The Token as an Instrument of Statecraft

There is a subtler security benefit. Digital Thorium creates a large, liquid, rupee-denominated asset class that non-aligned partner states can hold. Bhutan, Sri Lanka, Bangladesh, Nepal, and the African partners of the India Energy Alliance can receive Digital Thorium tokens in lieu of hard currency transfers. This turns energy into a unit of regional integration and weakens the dollar-denominated leverage of external actors. It is soft power with a capacity factor.

CHAPTER 6

Stakeholder Analysis

6.1 Who Wins, Who Pays, Who Waits

Policy viability depends on stakeholder arithmetic. A good policy makes most stakeholders better off, makes the losers politically absorbable, and the rest able to wait.

Stakeholder	What They Gain	What They Give Up
Retail saver	Access to a yield-bearing, AI-correlated, INR-denominated asset at an INR 500 ticket	Exposure to execution risk on the thorium build-out
Pension funds / insurers	Long-duration, inflation-linked, sovereign-backed instrument matching 30–60 year liabilities	Capital locked in a new asset class with thin secondary history for first five years
Hyperscalers (global)	20-year firm PPAs at known prices, clean attribute, proximity to the world's third largest AI user base	Must locate compute on Indian soil under Indian law
Indian AI companies	Cheap, firm, clean domestic compute at globally competitive cost	Higher immediate cost than subsidised foreign cloud
Government of India	Off-balance-sheet capital mobilisation, strategic autonomy, price discovery, export optionality	Some loss of direct operational control over the pace of the programme
DAE / BARC / IGCAR	Hard funding commitment, accelerated research budget, international credibility	Must accept market signals and external audit
NPCIL / BHAVINI	Capital at a lower cost of equity, faster build cadence	Must publish capacity factors and operating data
State governments (coastal)	Royalty share, industrial clustering around reactor sites, data centre investment	Local political cost of hosting nuclear infrastructure
Coal incumbents	Orderly transition via stranded-asset compensation bonds	Long-run market share
International partners	Access to a new strategic asset class; rupee-settlement instrument	Dollar-denominated leverage
Adversaries	Nothing	Leverage over Indian compute, electrons, and data

The retail saver is the politically decisive stakeholder. The entire architecture only works if 100–200 million households want to participate. Three features make this realistic:

1. **Price sensitivity.** An INR 500 minimum ticket matches the cheapest mutual fund SIP. Nothing in India Stack prevents this.
2. **Narrative legibility.** Owning electrons is easier to explain than owning derivatives. The story is: a rock from Odisha heats a reactor in Kalpakkam that powers a data centre in Chennai that serves an AI query answered in Patna, and the saver in Patna owns a slice of the rock.
3. **Religious and cultural resonance.** The Bhabha programme is already part of Indian science identity. Thorium is associated with national pride, not foreign technology.

The toughest stakeholder is the coal incumbent. The design response is explicit: a stranded-asset compensation bond that pays out a declining coupon to coal operators who retire capacity on a published schedule. This is cheaper than fighting them and makes the political transition possible.

6.2 Who Champions It

In the Indian political economy, a policy of this size needs at least one of: the Prime Minister's Office, the Finance Ministry, and the Department of Atomic Energy. Digital Thorium can be positioned as the signature economic instrument of the coming decade:

- For the PMO: the policy that delivered India's AI independence
- For the Finance Ministry: a non-inflationary source of infrastructure capital
- For the DAE: the mechanism that finally closed Bhabha's three-stage programme
- For SEBI: the flagship asset class of the next capital market decade
- For state governments on the coast: a decisive industrialisation vehicle

No single ministry owns it. That is a feature. The policy only works if the ministries coordinate, and the coordination itself improves long-run institutional quality.

CHAPTER 7

Risks and Mitigations

Honesty about failure modes is a precondition for market legitimacy. The dominant risks, with mitigations:

Risk	Mechanism	Mitigation
Reactor schedule slip	PFBR-class delays repeat across the fleet	Issue tokens only against commissioned capacity; use construction-stage warrants for earlier exposure
Capacity factor shortfall	Steady state below 82%	Token pays prorata; NTFC maintains reserve buffer
Speculative bubble in derivatives	Retail pile-in to leveraged products	Position limits, suitability tests, hard leverage caps on retail products
Proliferation narrative abroad	Foreign pushback on tokenised fissile assets	Public IAEA engagement; ledger auditable by designated observers
Cybersecurity of ledger	Exploit of the permissioned chain	Multi-sig custody, air-gapped cold storage for treasury, insurance pool
Political reversal	Future government unwinds the structure	Statutory basis; tokens held by 200M+ voters; sovereign bond-like protections
Retail loss event	First drawdown triggers political blowback	Deposit-insurance analogue capped at small ticket sizes
Foreign capture	Sovereign funds front-running retail	Tiered allocation; hard cap on aggregate foreign ownership per cohort
Displacement of renewables	Thorium financing crowds out solar/wind	Green taxonomy explicitly includes thorium; joint PPAs
Geopolitical sanctions	Secondary sanctions on Indian nuclear exports	Keep base-layer entirely domestic; export only downstream products

The single biggest risk is the first drawdown. Any retail market has its first bad year. Digital Thorium must enter its first bad year with a retail insurance mechanism, a clear narrative, and visible executive support. Without this, a transient 20 percent drawdown becomes a permanent political wound.

CHAPTER 8

Simulation: A Twenty-Year Path

To close the loop, here is a stylised forecast of the asset class under base, lift-off, and headwinds scenarios. All figures indicative and rounded.

Metric by 2046	Base	Lift-Off	Headwinds
Commissioned thorium capacity (GW)	50	85	18
Annual thorium generation (TWh)	359	610	130
Share of Indian electricity (%)	12	19	5
Share of Indian baseload (%)	22	34	8
Digital Thorium stock (INR lakh cr)	7.2	14	2.0
Retail participant households (million)	180	250	60
Fiscal benefit, annualised (INR lakh cr)	3.2	5.8	0.9
Oil import substitution ($B/yr)	22	38	6
Data centre capacity on thorium PPAs (GW)	30	55	6
Domestic AI workloads trained on Indian soil (%)	55	75	20

A few readings of this table are worth making explicit:

First, even the Headwinds scenario is better than the status quo. At 18 GW of thorium by 2046, India still gets double its current nuclear capacity, a non-trivial share of domestic AI compute, and the infrastructure for future acceleration. The asymmetry of outcomes favours the launch.

Second, the Lift-Off scenario is not utopian. It requires roughly 4.25 GW of thorium build per year, which is about the current annual rate of solar capacity addition. It is industrially feasible if the financing rails exist.

Third, the data centre capacity figures illustrate the strategic payoff. In the Base case, India ends up with the equivalent of one-third of current US data centre capacity, powered domestically, running on sovereign rupees. In Lift-Off, India becomes the second largest compute jurisdiction in the world.

CHAPTER 9

Policy Architecture

9.1 Statutory Skeleton

A compact legal stack is enough. Four instruments:

1. **The National Thorium Finance Act, 2027.** Creates the NTFC, defines the kWh-year unit, authorises retail issuance, caps foreign ownership, and places the statutory obligation on the Union to honour settlement.
2. **Amendment to the Atomic Energy Act, 1962.** Permits licensed private consortia to operate Stage 2 and Stage 3 reactors under DAE supervision, with capacity ring-fenced for NTFC tokenisation.
3. **SEBI Regulations on Energy Claims.** Defines Digital Thorium as a new regulated asset class, with disclosure standards, secondary market rules, and derivative oversight.
4. **Union Budget allocation for the NTFC backstop.** A contingent credit line sized at 0.1 percent of GDP, activated only on defined triggers.

9.2 Sequencing

1. **2027:** Statutory framework in place. NTFC incorporated. Pilot issuance against first 500 MW of PFBR-derived capacity.
2. **2028:** Retail rollout via India Stack integration. Secondary market goes live. First derivatives listed.
3. **2030:** 5 GW of tokenised thorium capacity. First hyperscaler PPA signed on token-referenced pricing.
4. **2035:** 20 GW commissioned. Digital Thorium is a mainstream asset class. Foreign sovereigns hold it in reserves.
5. **2046:** 50 GW commissioned under base case. India is structurally a compute exporter.

9.3 What To Avoid

Four temptations should be explicitly rejected:

- **Launching before the first reactor is commissioned.** Synthetic capacity would kill trust in the first drawdown.

- **Structuring it as a cryptocurrency.** The token is a claim on electrons, not on belief. The legal anchor must be Indian, the settlement rupee, the enforcement conventional.
- **Allowing leveraged retail products in the first five years.** Save that for after a full cycle of price discovery.
- **Letting foreign capital dominate the initial float.** The primary political legitimacy comes from Indian retail ownership. Foreign capacity can be added in tranches after year five.

CHAPTER 10

Closing

The core claim of this report is that three curves are about to cross. India's thorium reserve is the largest on earth. The world's appetite for electrons is about to vertical-take-off under the weight of AI training and inference. And the asset class preferences of an emerging middle class are tilting, with quiet historical inevitability, from land to energy.

When curves cross, policy matters. The default path leaves India with a state-run programme, financed from a thin budget, delivering a fraction of the potential, and watching the value of its own resource accrete to foreign cloud providers who buy the electrons once they exist. The alternative path, laid out here, is a market. A market turns the resource into a financeable asset class. It prices the electron. It prices the future. It lets a saver in Patna own a slice of the rock in Odisha that powers the query in Bengaluru that funds the startup in Mumbai.

This is not a conventional financial innovation. It is an ownership regime for the next century of Indian power. Land made the twentieth century. Electrons will make the twenty-first. Intelligence is what the electrons purchase. Digital Thorium is the instrument that lets India own the whole chain.

The capital is there. The reserve is there. The demand is arriving. What remains is the political act of building the rails.

Build them.

On sources and figures. *Reserve figures follow the Atomic Minerals Directorate and Department of Atomic Energy. Demand projections follow the IEA Energy and AI report (April 2025). The PFBR first-criticality date is 6 April 2026 per BHAVINI and IGCAR. Capex and LCOE bands are indicative, built from public OECD/NEA ranges for advanced reactors adjusted for Indian build cost. The simulation in Chapter 7 is illustrative and should be treated as such. No figure in this document should be used for investment purposes without independent verification.*